

METAL SOLUTIONS

EOS ToolSteel H13

Material Data Sheet

EOS TOOLSTEEL H13

High strength tool steel for hot and cold work tooling applications

EOS ToolSteel H13 is a versatile high strength tool steel for hot and cold work tooling applications. The high toughness and excellent thermal properties of H13 make it one of the most used material in hot work tooling. EOS ToolSteel H13 provides excellent properties to successfully bring AM into use for die casting production.

MAIN CHARACTERISTICS

- Chemical composition is similar to ASTM A681-08
- Excellent toughness and ductility
- Good high temperature strength
- High hot wear resistance

TYPICAL APPLICATIONS

- Die cast tooling
- Forging
- Hot extrusion tools

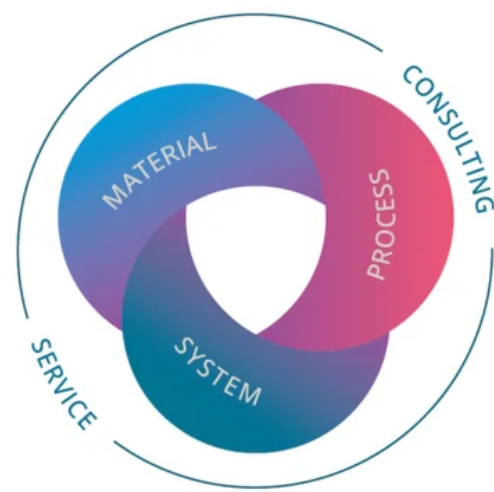
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

EOS ToolSteel H13 powder chemistry is according to ASTM A681-08.

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Fe		Balance
Cr	4.75	5.5
Mo	1.1	1.75
Si	0.8	1.25
V	0.8	1.2
Mn	0.2	0.6
C	0.32	0.45

Powder Particle Size

Generic Particle Size Distribution	15 - 63 µm
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HEAT TREATMENT

Description

EOS ToolSteel H13 can be heat treated to various hardness levels to match the needs of different applications. Heat treatment consists of three stages. First stage is a soft annealing process to relieve stresses from the built part. This treatment is performed for parts on the building platform. After soft annealing parts can be removed from the building platform and needed post processing and hardening heat treatment can be applied.

Steps

1. Soft annealing

Heating with moderate heating rate and then hold 30 min at 850 °C (± 10 °C) when parts have thoroughly heated through. Next cool slowly to 650 °C with a cooling rate of 10 °C/ hour. After reaching 650 °C cooling in air or with low pressure gas cooling.

2. Hardening

Heating with moderate heating rate and then hold 30 min at 1 045 °C (± 10 °C) when parts have thoroughly heated through. Quench the parts with high speed gas cooling or warm oil. Optimally continue directly to tempering when reaching 50-70 °C.

3. Tempering

2 times 2 h at 580-650 °C depending on the desired hardness level.

Hold for 2 h at desired temperature when parts have thoroughly heated through. Cool to room temperature between temperings.

EOS ToolSteel H13 for EOS M 290 | 40 µm

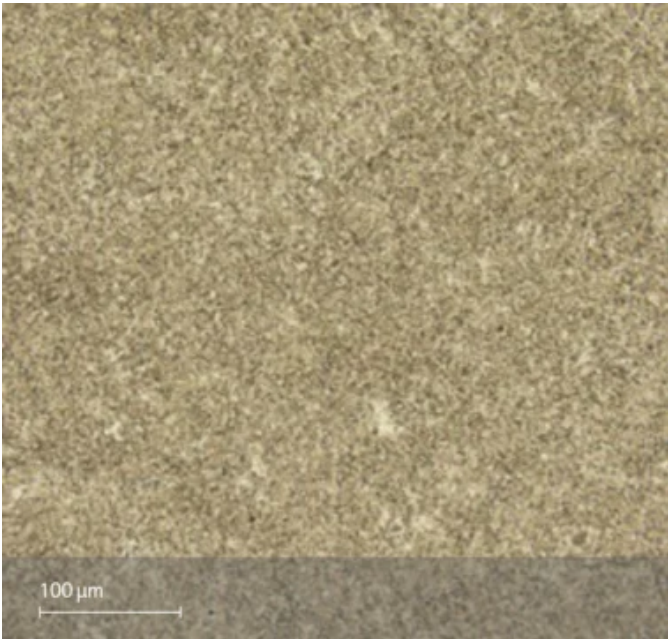
EOS M 290 - 40 µm - TRL 3

System Setup	EOS M 290
EOS Material set	H13_040_CoreM291
Software Requirements	EOSPRINT 2.6.21 or newer EOSYSTEM 2.10.65 or newer
Recoater Blade	Ceramic
Nozzle	EOS Grid Nozzle
Inert gas	Argon, Nitrogen
Sieve	90 µm

Additional Information	
Layer Thickness	40 µm
Volume Rate	3.7 mm³/s

Chemical and Physical Properties of Parts

Chemical composition of built parts matches the chemistry of EOS ToolSteel H13 powder.



Micrograph etched, heat treated state. Etchant: Marble

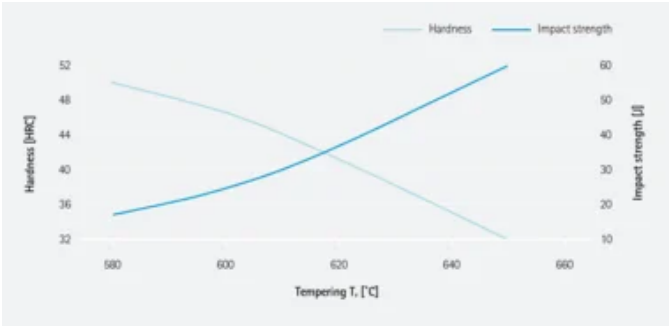
Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 μm	0.08 %	-

Mechanical Properties

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	1490	1680	9	-	-	-
Horizontal	1480	1670	9	-	-	-



Achieved hardness and impact toughness in relation to the tempering temperature. Hardness tested as per ISO 6508-1 and impact toughness tested as per ISO 148-1, V-notch, room temperature.

Heat treated to 50 HRC

EOS ToolSteel H13 can be heat treated to various hardness levels to match the needs of different applications. Heat treatment consists of three stages. First stage is a soft annealing process to relieve stresses form the built part. This treatment is to performed for parts on the building platform. After soft annealing parts can be removed from the building platform and needed post processing and hardening heat treatment can be applied.

1. Soft annealing

Heating with moderate heating rate and then hold 30 min at 850 °C (±10 °C) when parts have thoroughly heated through. Next cool slowly to 650 °C with a cooling rate of 10 °C/ hour. After reaching 650 °C cooling in air or with low pressure gas cooling.

2. Hardening

Heating with moderate heating rate and then hold 30 min at 1 045 °C (±10 °C) when parts have thoroughly heated through. Quench the parts with high speed gas cooling or warm oil. Optimally continue directly to tempering when reaching 50-70 °C.

3. Tempering

2 times 2 h at 580-650 °C depending on the desired hardness level.

Hold for 2 h at desired temperature when parts have thoroughly heated through. Cool to room temperature between temperings.

Please note:

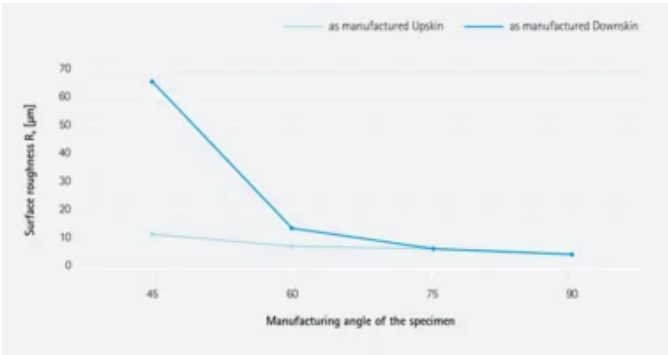
Due to high stress build up consideration on part size and shape is needed. Parts with 150 mm cross-section can be reliably built with good design principles and when instructions are followed.

Multiple smaller parts can be built on the same platform. To control temperature a 10 mm perforated hatch support is to be applied for each part unless when building a hybrid part on H13 base. For hybrid parts, size of the H13 base needs consideration due to limited heating capacity of hardware.

A soft annealing heat treatment (stage 1) is to be applied to parts on building platform before any manipulation, post processing or removal from platform.

After soft annealing parts can be heat hardened and tempered according to instructions and post processed as conventional H13 parts.

Surface Roughness



Surface roughness

Coefficient of Thermal Expansion

ASTM E228	Temperature
11.3*10 ⁻⁶ /K	25 – 100 °C
11.7*10 ⁻⁶ /K	25 – 200 °C
12.1*10 ⁻⁶ /K	25 – 300 °C
12.5*10 ⁻⁶ /K	25 – 400 °C
12.8*10 ⁻⁶ /K	25 – 500 °C
13*10 ⁻⁶ /K	25 – 600 °C

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Status as of 18.08.2026. Subject to technical modifications. EOS is certified according to ISO 9001.

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